

Moving-Knife Bisection: A Directional Reock-Scored Splitter for BISECT

T.13 — Algorithm Design / Geometric Structure

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Abstract

MKA is BISECT’s directional sweep splitter for Reock-scored bisection. At each bisection-tree node it projects tract centroids into Albers EPSG:5070, tests n_{orient} orientations in $[0, \pi)$, sorts tracts by projection onto each sweep axis, cuts at the first prefix whose population reaches one half of the node population, scores the two halves by the minimum of their centroid-MEC Reock approximations, and keeps the best orientation. The selected assignment is then passed through the same bounded local boundary rebalance family used by the other geometric splitters.

This paper corrects the earlier presentation of Moving-Knife. The implemented algorithm is a recursive two-way bisection primitive, not a proof of global Reock optimality for a final k -district plan. Its current Reock score is a centroid approximation: the denominator is the minimum enclosing circle of projected tract centroids, and the numerator falls back to the number of centroids in the current implementation. The `polsby-popper` metric path currently falls back to the same Reock scorer because perimeter data are not yet available in the split routine. The method is nevertheless useful: it gives BISECT a legible geometric baseline whose candidate choices can be audited as “show the sweep, show the population prefix, show the worst-half score.”

The publication gap is empirical and archival. NC/FL/WA quality tables, partisan-neutrality claims, per-node optimal-angle manifests, polygon-boundary Reock, and

true Polsby-Popper scoring all require archived run packages before they should carry certification weight. Implementation: `-structure moving-knife`.

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1. Introduction

1.1. The Splitter We Need

BISECT has several ways to split a graph node. METIS follows graph boundary structure. CVD-GEO assigns tracts to geographic centers. BFGGROWTH grows two regions from seeds. MKA asks a different question:

Which straight sweep direction gives the best worst-half Reock score after the node is divided into two population-balanced prefixes?

The question is intentionally local. BISECT builds a final plan by recursive bisection, so MKA is not a global optimizer over all k districts. It is a node-level structure primitive: take the current node, try many sweep angles, choose the best two-way split under the implemented score, then continue down the bisection tree.

1.2. Show the Sweep, Not the Slogan

The moving-knife idea is easy to say and easy to overstate. The useful visual object is not merely a label such as “47 degrees.” It is the ordered prefix created by projecting every tract onto the candidate axis.

Horizontal sweep

1	2	3	4
1	2	3	4
1	2	3	4

Diagonal sweep

1	1	2	3
1	2	3	4
2	3	4	4

Vertical sweep

1	1	1	1
2	2	2	2
3	3	3	3

The numbers are projection ranks. For each angle, the algorithm reads tracts from rank 1 upward until the cumulative population reaches half of the node. That prefix becomes one side of the candidate split; the remaining suffix becomes the other side. The selected angle is the one whose two halves have the largest minimum Reock score under the current centroid-MEC approximation.

1.3. Why Reock Appears Here

Reock compactness compares a district’s area to the area of the smallest circle that encloses it [Reock, 1961]. It is a natural score for a sweep-based splitter because bad directions create long, thin prefixes whose enclosing circle grows quickly.

Short enclosing circle Long enclosing circle

L	L	R
L	L	R
R	R	R

L	L	L	L	R
R	R	R	R	R

The left candidate is blocky. The right candidate stretches the left half across most of the state, so its minimum enclosing circle is large relative to its area. MKA prefers the former if population balance permits it.

1.4. Implementation Status

The implementation entry points live in the bisection runner:

```
split_subgraph_mka   split_subgraph_mka_direction   run_all_splits_mka
```

The current contract is:

- centroid data are required; missing centroids produce a configuration error rather than a silent fallback;
- coordinates are projected with the same Albers helper used by geographic CVD;
- the default metric is Reock, computed from centroid points;
- `polsby-popper` is parsed, but the current split routine falls back to Reock because perimeter data are not present there;
- the post-sweep boundary rebalance is local and bounded, so final validation remains a required downstream step.

1.5. Contributions

This paper contributes an implementation-aligned account of MKA:

1. It explains the actual two-way recursive bisection routine and its orientation scan.
2. It gives visual checks for projection rank, population prefix, and worst-half Reock scoring.

3. It separates useful fair-division language from legal or empirical claims that require archived evidence packages.

The result is a practical structure-layer paper, not a certification brief. It shows what MKA currently does, why the choices are inspectable, and what must be added before the method can support stronger publication claims.

2. Background

2.1. Moving-Knife as a Procedural Metaphor

Fair-division theory studies procedures for dividing a resource among claimants. Dubins and Spanier’s moving-knife protocol is one of the classic continuous procedures [Dubins and Spanier, 1961]: a knife moves across the resource, and the stopping rule determines the allocation.

BISECT’s MKA does not import all of the envy-free or proportionality guarantees from cake-cutting. Redistricting has fixed geography, indivisible tracts, population balance constraints, contiguity requirements, and legal criteria that do not map cleanly onto private valuations. The useful transfer is procedural: a public, mechanical sweep is easier to explain and audit than an opaque choice among many candidate maps.

2.2. Reock Compactness

Reock compactness is

$$\text{Reock}(D) = \frac{A(D)}{A(\text{MEC}(D))}, \quad (1)$$

where $\text{MEC}(D)$ is the minimum enclosing circle of district D [Reock, 1961]. A compact round region fills much of its enclosing circle; a long thin region does not.

The implemented scorer is an approximation. It computes the minimum enclosing circle of projected tract centroids, not the exact district polygon boundary. When explicit tract areas are not passed to the scorer, the numerator is the number of centroid points. The resulting value is clamped to $[0, 1]$.

Remark 1 (Current score versus publication Reock). *The current split scorer is useful for ranking candidate sweep directions inside the algorithm. It is not yet a final legal Reock*

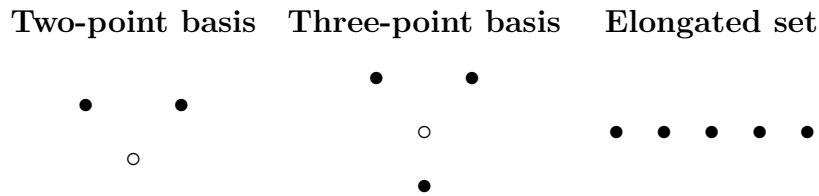
Table 1: Structure-layer roles. “Direct target” means the implemented split routine scores candidates by that quantity; it does not imply final-plan global optimality.

Algorithm	Candidate mechanism	Direct target	Main caveat
METIS	graph partition	edge cut	graph objective, not Reock
CVD-GEO	geographic assignment	seed proximity	half-split target today
BFSGROWTH	two-frontier growth	legible contiguity	baseline, not compactness proof
MKA	projection sweep	centroid-MEC Reock	local and approximate

statistic for a certified district plan. A publication-grade Reock package should compute the minimum enclosing circle of polygon boundary vertices and use true district area in the numerator.

2.3. Minimum Enclosing Circles

The implementation uses a deterministic-shuffle version of Welzl’s randomized incremental minimum-enclosing-circle algorithm [Welzl, 1991]. For a set of points in the plane, the enclosing circle is determined by at most three boundary points. This makes the scoring routine fast enough to call twice per candidate orientation.



The third sketch is the danger case MKA tries to avoid: a sweep that creates an elongated half has a large enclosing circle and therefore a lower score.

2.4. Where MKA Sits in the Portfolio

MKA is therefore best understood as a diagnostic and construction option for Reock-shaped geometry. It supplies a clear candidate ledger: orientation, population prefix, left score, right score, selected minimum. That ledger is the bridge to future audit artifacts.

3. The Moving-Knife Algorithm

3.1. Notation

Let $G_v = (V_v, E_v)$ be the current bisection-tree node with $m = |V_v|$ tracts. Each tract i has population p_i and centroid $\ell_i = (\lambda_i, \phi_i)$. The implementation projects centroids to Albers coordinates $\mathbf{x}_i = (x_i, y_i)$ before scoring directions.

3.2. The Implemented Split

For each orientation $\theta_k = k\pi/n_{\text{orient}}$, MKA computes the projection

$$q_i(\theta_k) = x_i \cos \theta_k + y_i \sin \theta_k.$$

Sorting by q_i turns the node into a one-dimensional queue. The split point is the first prefix whose cumulative population reaches $P_v/2$.

rank	1	2	3	4	5	6	7	8	candidate
pop	4	3	2	6	1	5	2	3	$P_v = 26$
cum	4	7	9	15	16	21	23	26	stop at 15
side	L	L	L	L	R	R	R	R	L/R

The score for this candidate is

$$\min(\text{Reock}_{\text{centroid}}(L), \text{Reock}_{\text{centroid}}(R)).$$

The algorithm keeps the assignment with the largest score across all tested orientations.

3.3. Seed Derivation

The code defines a deterministic node seed:

$$s_v^{\text{mka}} = \text{first8}(\text{SHA-256}(\text{"MKA_INIT_"} \parallel \text{le4}(|\text{node_path}|) \parallel \text{node_path} \parallel \text{le8}(\text{base_seed}))).$$

The prefix is tested to be distinct from CVD seed prefixes. In the current split routine this value is computed as a reserved tie seed, but ties are resolved by the first strictly best

Algorithm 1 MKA-BISECT($G_v, \mathbf{x}, \mathbf{p}, n_{\text{orient}}, \tau$)

```
1: sort  $V_v$  by global tract id for deterministic local indexing
2: build local adjacency for the later boundary rebalance
3:  $P \leftarrow \sum_{i \in V_v} p_i$ ;  $P^* \leftarrow \lfloor P/2 \rfloor$ 
4: project all available centroids to Albers coordinates
5: best_score  $\leftarrow -\infty$ ; best_assignment  $\leftarrow$  all right
6: for  $k = 0$  to  $n_{\text{orient}} - 1$  do
7:    $\theta \leftarrow k\pi/n_{\text{orient}}$ 
8:   compute  $q_i = x_i \cos \theta + y_i \sin \theta$  for every local tract  $i$ 
9:   sort tracts by  $q_i$ 
10:  scan the sorted list until cumulative population reaches  $P^*$ 
11:  assign the prefix left and the suffix right
12:  compute centroid-MEC Reock for left and right
13:  score  $\leftarrow \min(r_L, r_R)$ 
14:  if score  $>$  best_score then
15:    keep this assignment
16:  end if
17: end for
18: rebalance for at most 200 boundary moves toward  $P/2$ 
19: repair degenerate empty-side cases if necessary
20: return left and right global tract sets
```

orientation encountered rather than by a random draw. The seed formula should still be retained in manifests once per-node angle ledgers are archived.

3.4. Pseudocode

3.5. Boundary Rebalance

The sweep can overshoot the half-population target by one tract. After the best orientation is chosen, the code repeatedly moves a boundary tract from the heavier side to the lighter side when that move best reduces the population excess. This loop is capped at 200 iterations.

The loop is deliberately local. It considers only tracts adjacent to the other side, but it does not run a full contiguity proof after every tentative move. As with the other structure splitters, the returned plan should be validated by the downstream plan checks before any certification claim is made.

3.6. Data and CLI

MKA requires tract centroids. If centroids are absent, the CLI returns a configuration error:

[CONFIG] --structure moving-knife requires centroid data.

Run: `bisect fetch --type centroids`

The standard invocation is:

```
bisect state --state NC --year 2020 \  
  --structure moving-knife \  
  --mka-orientations 180 \  
  --mka-metric reock
```

`--mka-metric polsby-popper` is accepted by the enum, but currently falls back to the Reock scoring path in `split_subgraph_mka`.

3.7. Runtime

Each orientation requires $O(m)$ projection work, $O(m \log m)$ sorting, a linear population scan, and two minimum-enclosing-circle scores. The resulting node-level runtime is

$$O(n_{\text{orient}} m \log m + 200m).$$

For the default $n_{\text{orient}} = 180$, this is heavier than one simple graph split but still small enough to serve as an auditable geometric baseline.

4. Comparison and Evidence Plan

4.1. Current Evidence Status

The implementation has unit coverage for parsing, deterministic behavior, seed-prefix separation, direction selection on simple grids, empty-centroid failure, and full `run_all_splits_mka` smoke behavior. That is enough to trust the algorithmic path for development. It is not enough to publish statewide comparative performance claims.

4.2. How the Benchmark Should Look

The old version of this paper reported concrete NC/FL/WA numbers. Those numbers should be treated as placeholders until the run directories, inputs, seeds, manifests, and score scripts are archived. The correct benchmark table has this shape:

Table 2: Evidence status for Moving-Knife claims.

Claim	Status	Required next artifact
CLI parses moving-knife	implemented	keep regression tests
Centroids are required	implemented	manifest error examples
Orientation sweep is deterministic	implemented	per-node angle ledger
Centroid-MEC Reock ranking works	implemented	replayable trace package
Polygon-boundary Reock	not implemented here	boundary-vertex scorer
Polsby-Popper MKA	enum only	perimeter-aware split scorer
NC/FL/WA superiority	target claim	archived same-metric runs
Partisan neutrality	target claim	multi-seed outcome packages

For publication, the package should record:

- input graph and centroid hashes;
- base seed and bisection tree node paths;
- `mka_orientations` and metric;
- for every split node, the selected angle and score;
- final plan assignments and validation results;
- identical scoring scripts for all compared algorithms.

4.3. A Worked Candidate Ledger

The key reader-facing evidence is a candidate ledger. A small node can be shown as follows:

θ	prefix pop	left shape	right shape	min score	decision
0°	51.2%	long	compact	0.31	reject
45°	50.6%	blocky	blocky	0.44	keep
90°	52.0%	compact	long	0.29	reject
135°	49.8%	split tail	blocky	0.37	reject

This is the Moving-Knife equivalent of “show the split, not say the split.” The reader can see why one direction wins: it gives both halves acceptable enclosing-circle geometry.

Table 3: Moving-Knife benchmark package target. All entries are pending archived same-metric runs.

State	Algorithm	Min centroid-Reock	Mean centroid-Reock	EC	Package
NC	METIS	TBD	TBD	TBD	TBD
NC	CVD-Geo	TBD	TBD	TBD	TBD
NC	BFSGROWTH	TBD	TBD	TBD	TBD
NC	MKA	TBD	TBD	TBD	TBD
FL	METIS	TBD	TBD	TBD	TBD
FL	MKA	TBD	TBD	TBD	TBD
WA	METIS	TBD	TBD	TBD	TBD
WA	MKA	TBD	TBD	TBD	TBD

4.4. Orientation Granularity

The default 180-orientation sweep tests one-degree increments. Smaller sweeps such as 36 orientations can be useful for development, but production claims should report the granularity explicitly. The benchmark question is not merely runtime. It is whether coarser scans select the same bisection-tree angles and produce comparable final validation metrics.

4.5. Partisan and Legal Metrics

MKA does not read election returns in the split routine. That makes the structure step formally independent of partisan outcome data, but it does not prove partisan neutrality of final plans. Recursive bisection, geography, and population distribution can still create predictable electoral effects. Publication claims should therefore compare final plans across seeds using the same partisan metrics used elsewhere in BISECT.

5. The Fair-Division Argument

5.1. What the Argument Can Say

The fair-division value of MKA is procedural intelligibility. A commission can explain the structure step in ordinary geometric language:

We rotated a public set of candidate sweep directions across the current region.

For each direction, the split point was fixed by population balance. We scored

both halves by the same compactness rule and chose the direction whose weaker half scored best.

That statement is accurate for the implemented node-level algorithm. It is also auditable: an outside reviewer can replay the projection order and cumulative population table for each angle.

5.2. What the Argument Cannot Say Yet

MKA should not be described as a complete legal certificate. Several boundaries matter:

- it optimizes a local bisection score, not the global Reock score of all final districts;
- current scoring uses centroid points rather than full polygon boundary vertices;
- the local rebalance can alter the selected sweep assignment;
- VRA, community, nesting, county, and partisan fairness criteria are outside the split objective;
- final-plan claims require archived assignments and independent scoring.

These limits do not weaken MKA as a construction primitive. They make its publication boundary clear.

5.3. Court-Facing Language

The strongest court-facing formulation is modest:

The Moving-Knife run used a predetermined, geography-only sweep procedure to generate each recursive split. The selected split at each node maximized the minimum of the two implemented Reock-approximation scores among the tested orientations. The run package records enough information for an independent reviewer to reproduce the selected orientations and verify the final plan against statutory criteria.

Once per-node angle manifests and polygon-boundary Reock are implemented, this language can be strengthened. Until then, the paper should emphasize auditability and algorithmic neutrality, not legal sufficiency.

5.4. Relationship to METIS and CVD

METIS can be explained as minimizing graph boundary. CVD-GEO can be explained as assigning tracts to geographic centers. MKA can be explained as rotating a ruler across the map and choosing the split that best avoids a weak, elongated half. The last explanation is unusually visual, which is why this paper is useful even before the certification package is complete.

6. Conclusion

MKA gives BISECT a simple directional sweep alternative to graph cuts, Voronoi assignment, and BFS growth. It is visually legible: for every candidate angle, the reader can inspect the projection order, population prefix, two halves, and worst-half Reock approximation.

Use today. Use `-structure moving-knife` when a run needs a geometry-first baseline whose split choices are easy to replay. Treat the output as a candidate structure plan that still needs the standard validation and scoring pipeline.

Do not overclaim. The current implementation does not prove final-plan Reock optimality, does not compute polygon-boundary Reock, and does not implement true Polsby-Popper optimization inside `split_subgraph_mka`. It is a credible local splitter, not yet a standalone certification system.

Next work.

1. Archive per-node angle ledgers: projection order, prefix population, left/right scores, selected angle, and post-rebalance changes.
2. Add polygon-boundary Reock scoring for final plans and, if feasible, for candidate split ranking.
3. Implement real Polsby-Popper scoring only when perimeter data are present in the split routine.
4. Run same-metric NC/FL/WA packages against METIS, CVD-GEO, and BFSGROWTH.

5. Add visual atlas panels that show each rejected angle and why the winner has the best worst-half enclosing-circle score.

Summary. The useful contribution is not the phrase “moving knife.” It is the visible ledger: show the sweep, show the prefix, show both enclosing-circle scores, and show why that split was chosen. That is the standard this paper now sets for MKA.

References

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